THE SPHERE PACKING PROBLEM IN DIMENSION 7

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Abstract. Every Voronoi cell of a unit-ball packing of $\mathbb{R}^7$ has volume greater than 14.32, and some cells have volume $232/15 = 15.466\ldots$, less than the volume 16 of the cell of $\sqrt{2}\,E_7$; around the full kissing configuration of E_7 this is the exact minimum. The small cells come from extra balls placed beyond the fifty-six deep holes of E_7, and the construction applies to a lattice exactly when its Voronoi cell is not contained in the convex hull of its minimal vectors. Consequently the density $\pi^3/105$ of E_7 cannot be proved cell by cell; whether it is the largest density in dimension 7 remains open.

1. Introduction

A *unit-ball packing* of $\mathbb{R}^n$ is a set $\mathcal{C} \subset \mathbb{R}^n$ of centres any two of which are at distance at least 2. Its *density* is the upper limit, as $R \to \infty$, of the fraction of the ball of radius R covered by the unit balls around the centres, and Δ_n is the supremum of these densities. The value of Δ_n is known in dimensions 1, 2, 3, 8 and 24 only: the hexagonal lattice in the plane (Thue, Fejes Tóth [FT72]), the face-centred cubic lattice in space (Hales [Hal05, HAB+17]), E_8 (Viazovska [Via17]) and the Leech lattice (Cohn, Kumar, Miller, Radchenko and Viazovska [CKMRV17]). In dimension 7 the densest known packing is the lattice E_7, scaled so that its minimal vectors have length 2; its density is

$$\Delta(\sqrt{2}\,E_7) = \frac{\mathrm{vol}(B^7)}{16} = \frac{\pi^3}{105} = 0.2952\ldots, \tag{1.1}$$

where B^7 is the unit ball and 16 is the volume of a Voronoi cell of $\sqrt{2}\,E_7$. It is the densest lattice packing in dimension 7 (Blichfeldt [Bli35]). For arbitrary packings the linear programming method of Cohn and Elkies gives the upper bound 0.3276 [CE03, Table 3], and the gap to (1.1) is about eleven per cent.[1]

1.1. The local approach.

The oldest way to bound the density is local. The Voronoi cell of a centre c,

$$V_c = \{z \in \mathbb{R}^n : \ |z - c| \le |z - c'| \text{ for all } c' \in \mathcal{C}\},$$

contains the unit ball around c, and the cells of the different centres tile $\mathbb{R}^n$ with disjoint interiors. If one could show that every cell in every packing has volume at least some number ω_n, the density would be at most $\mathrm{vol}(B^n)/\omega_n$. For a lattice packing all cells are translates of one polytope of volume equal to the covolume of the lattice, so the best one can hope for from this argument is $\omega_n =$ covol of the densest lattice, and the argument would then prove that this lattice is optimal among all packings. In dimension 3 this hope is dashed by the regular dodecahedron: a cell can be smaller than the rhombic dodecahedron of the face-centred cubic

Date: November 2026.

2020 *Mathematics Subject Classification.* Primary 52C17; Secondary 11H31, 52B11, 52C07.

Key words and phrases. Sphere packing; Voronoi cell; E_7 lattice; deep hole; Rogers bound.

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Versions 1–9 of this paper (Preprints.org, doi:10.20944/preprints202606.1534.v9 and earlier) asserted, as a theorem, that every Voronoi cell of a unit-ball packing of $\mathbb{R}^7$ has volume at least 16, and deduced from it that the density of such packings is at most $\pi^3/105$. Theorem 1.1 below shows that the first assertion is false, so the deduction has no content, and Section 8 locates the step of the earlier argument that fails. The present version withdraws both claims and replaces them by what is proved here.

[1] In [CE03, Table 3] the bounds are given as centre densities, $\Delta/\mathrm{vol}(B^7)$: 0.0625 for $\sqrt{2}\,E_7$, 0.06933 for the linear programming bound and 0.06981 for Rogers' bound.

lattice, which is why the proof of the Kepler conjecture needs weighted averages of cells rather than a bound on each cell (Hales [Hal05, Hal12]; the dodecahedral conjecture itself was proved by Hales and McLaughlin [HM10]). In dimension 4 the corresponding local statement, that every cell has volume at least that of the 24-cell, is Musin's 24-cell conjecture [Mus18], which is open. In dimension 7 the local statement would read: *every Voronoi cell of a unit-ball packing of* $\mathbb{R}^7$ *has volume at least* 16. Earlier versions of this paper asserted that statement. It is false.

1.2. **Results.** Write $\Lambda = \sqrt{2}\,E_7$, let M be the set of its 126 minimal vectors (each of length 2) and P its Voronoi cell at the origin, a polytope of volume 16 with 126 facets and 632 vertices. The vertices of P are the holes of the lattice; 56 of them are the deep holes, at distance $\sqrt{3}$ from the origin, and each deep hole lies on exactly 27 facets. Fix a deep hole v, write $u = v/\sqrt{3}$ for its direction and put

$$y_r = r\,u \qquad \text{for } \tfrac{4}{\sqrt{3}} \le r \le 2\sqrt{3}.$$

Theorem 1.1 (One extra centre)**.** *The set* $\mathcal{C}_r = \{0\} \cup M \cup \{y_r\}$ *is a unit-ball packing of* $\mathbb{R}^7$*; the extra centre* $y_{4/\sqrt{3}}$ *touches* 27 *of the* 126 *balls around the minimal vectors. The Voronoi cell of the origin in* $\mathcal{C}_r$ *has volume*

$$\operatorname{vol} V_0(\mathcal{C}_r) = 16 - \frac{1}{105}\Big(3 - \frac{\sqrt{3}}{2}\,r\Big)^7, \tag{1.2}$$

which at $r = 4/\sqrt{3}$ *equals* $16 - \frac{1}{105}$*. The same formula holds for the cell of the origin in every packing that contains* $\mathcal{C}_r$ *and whose remaining centres are at distance at least* $2\sqrt{3}$ *from the origin, and every packing containing* $\mathcal{C}_r$ *has a cell at the origin of volume at most the right-hand side of* (1.2)*.*

The fifty-six deep holes can be cut simultaneously.

Theorem 1.2 (All deep holes)**.** *Let* $y_1, \dots, y_{56}$ *be the points* $(4/\sqrt{3})\,v_i/\sqrt{3}$ *for the fifty-six deep holes* v_i *of* P*. Then* $\{0\} \cup M \cup \{y_1, \dots, y_{56}\}$ *is a unit-ball packing of* $\mathbb{R}^7$ *with* 183 *centres, any two of the* y_i *are at distance at least* $8/3$*, and the Voronoi cell of the origin has volume*

$$16 - \frac{56}{105} = \frac{232}{15} = 15.4666\ldots$$

Packings with finitely many centres are legitimate packings, but one may prefer saturated ones, where no further ball can be added, or periodic ones. Both are available.

Corollary 1.3. *There is a periodic, saturated unit-ball packing of* $\mathbb{R}^7$ *in which the Voronoi cell of some centre has volume at most* $232/15$*. Consequently no lower bound on the volume of a single Voronoi cell can prove that* $\Delta_7 = \pi^3/105$*.*

How small can a cell be? Let

$$\omega_7 = \inf\{\operatorname{vol} V_c(\mathcal{C}) :\ \mathcal{C} \text{ a unit-ball packing of } \mathbb{R}^7,\ c \in \mathcal{C}\}.$$

The simplex bound of Rogers, in the cell-by-cell form of Böröczky, bounds ω_7 from below, and together with Theorem 1.2 it confines ω_7 to a window that lies strictly below 16.

Theorem 1.4 (The window for the smallest cell)**.**

$$14.3234 \;<\; \frac{1}{1260\,\Omega_7} \;\le\; \omega_7 \;\le\; \frac{232}{15} = 15.4666\ldots \;<\; 16,$$

where $\Omega_7 = 5.54093\ldots \times 10^{-5}$ *is the fraction of the full solid angle subtended at a vertex of a regular 7-simplex. In particular every Voronoi cell of every unit-ball packing of* $\mathbb{R}^7$ *has volume greater than* 14.3234*, so the ball fills at most* $32.99\,\%$ *of its cell.*

The upper end of the window is sharp for every packing in which a ball is surrounded by the full kissing configuration of E_7.

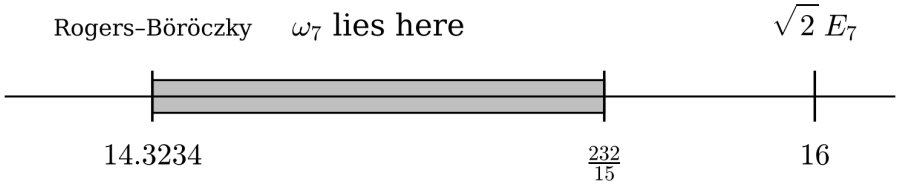


FIGURE 1. Where the smallest Voronoi cell of a unit-ball packing of $\mathbb{R}^7$ can lie (Theorem 1.4); the cell of $\sqrt{2}\,E_7$ is outside the window.

Theorem 1.5 (The E_7 shell)**.** *Let $\mathcal{C}$ be a unit-ball packing of $\mathbb{R}^7$ that contains $\{0\} \cup M$. Then*

$$V_0(\mathcal{C}) \;\supseteq\; P \cap \operatorname{conv}(M) \qquad \text{and} \qquad \operatorname{vol} V_0(\mathcal{C}) \;\geq\; \operatorname{vol}(P \cap \operatorname{conv}(M)) = \frac{232}{15},$$

with equality for the packing of Theorem 1.2. Thus $232/15$ is the smallest volume of a Voronoi cell whose centre has the 126 neighbours of $\sqrt{2}\,E_7$.

The proof (Section 5) rests on one line: an admissible centre y outside $\{0\} \cup M$ satisfies $\langle m, y\rangle \leq |y|^2/2$ for every $m \in M$, and this inequality is linear in m. Whether $\omega_7 < 232/15$, that is, whether a cell can be made smaller by moving the neighbours themselves, we do not know.

The mechanism is general and has a clean criterion. Let $\Lambda \subset \mathbb{R}^n$ be a lattice with minimal distance 2, with Voronoi cell P and covering radius ρ, and suppose that the Voronoi-relevant vectors of Λ are its minimal vectors M (this holds for all the root lattices).

Theorem 1.6 (The one-point criterion)**.** *Let u be a unit vector. There is a point y on the ray $\mathbb{R}_{>0}u$ such that $\{0\} \cup M \cup \{y\}$ is a packing and $\operatorname{vol} V_0(\{0\} \cup M \cup \{y\}) < \operatorname{vol} P$ if and only if the support function of P exceeds that of $\operatorname{conv}(M)$ at u. In particular no single extra centre, in any direction, shrinks the cell below $\operatorname{vol} P$ if and only if $P \subseteq \operatorname{conv}(M)$, and this inclusion holds if and only if $\rho \leq \sqrt{2}$. If $\rho > \sqrt{2}$, the extra centre $\max(2, 4/\rho)\, v/|v|$ in the direction of any deep hole v shrinks the cell.*

Theorem 1.7 (Root lattices)**.** *Among the irreducible root lattices scaled to minimal distance 2, the inclusion $P \subseteq \operatorname{conv}(M)$ holds for A_1, A_2, A_3, D_4 and E_8 and fails for A_n with $n \geq 4$, for D_n with $n \geq 5$, for E_6 and for E_7. The volume removed by the deep-hole centre of Theorem 1.6, as a fraction of $\operatorname{vol} P$, is*

Λ	A_4	A_5	D_5	D_6	D_7	E_6	E_7
fraction removed	$\frac{1}{1440}$	$\frac{1}{240}$	$\frac{1}{2880}$	$\frac{1}{576}$	$\frac{817}{201600}$	$\frac{1}{3240}$	$\frac{1}{1680}$

The lattices that pass the criterion are the hexagonal lattice, the face-centred cubic lattice, D_4 and E_8 (and $\mathbb{Z}$): precisely the lattices of the dimensions where the sphere packing problem is solved, together with the lattice of Musin's conjecture.[2] Passing the criterion is necessary for the local approach and not sufficient: A_3 passes it, and still the Voronoi cells in dimension 3 can be smaller than the rhombic dodecahedron. In dimension 7 the criterion fails, so the local approach does not even get off the ground, and Theorem 1.2 says it fails by a wide margin.

1.3. **What is derived here and how.** Theorem 1.4 is proved in Section 6: the lower bound is the Rogers–Böröczky inequality, and the only number to compute is the solid angle Ω_7, which is the orthant probability of a Gaussian vector with correlations $-\frac{1}{7}$.

Theorem 1.5 is proved in Section 5 from one inequality and polar duality, with one exact vertex enumeration. Everything in Theorems 1.1, 1.2 and 1.6 is proved in the text by hand up to two finite computations in exact rational arithmetic: the enumeration of the vertices of a polytope given by its facets (the double description method) and the volume of a polytope.

[2]The Leech lattice also has covering radius $\sqrt{2}$ times its packing radius (Conway, Parker and Sloane [CPS82]), but its relevant vectors are not all minimal, so Theorem 1.6 does not apply to it as stated.

Section 4 derives the structure of P at a deep hole, the cone of 27 facets and its cross-section, a scaled copy of the polar of the Schläfli polytope 2_{21}, and shows that the cap removed by the extra centre is a truncation of that cone, which gives the seventh power in (1.2); the single number $\frac{1}{105}$ is then an exact computation, done twice in independent ways. Theorem 1.7 follows from the covering radii for the lattices that pass and from the same exact computation for the others. Section 8 identifies the step of the earlier versions that fails. Section 10 describes the computations and the accompanying programs, in Julia and in Lean. Section 9 discusses what remains of the sphere packing problem in dimension 7 after these results.

1.4. **Conventions.** Throughout, $\langle \cdot, \cdot \rangle$ is the standard inner product, $|x|$ the Euclidean norm, B^n the closed unit ball, vol the Lebesgue measure of the ambient dimension unless a subscript says otherwise, and $h_K(u) = \max_{z \in K} \langle z, u \rangle$ the support function of a compact convex set K. Lattices are always scaled so that their minimal vectors have length 2, so that the lattice points are the centres of a unit-ball packing; when we need the usual normalisation of a root lattice, with roots of squared length 2, we say so.

2. Voronoi cells, holes and the shell bound

2.1. **Voronoi cells of packings.** Let $\mathcal{C}$ be a unit-ball packing and $c \in \mathcal{C}$. The cell V_c is the intersection of the closed half-spaces

$$H_c(c') = \{z : \ \langle z - c,\, c' - c \rangle \le \tfrac{1}{2}|c' - c|^2\}, \qquad c' \in \mathcal{C} \setminus \{c\},$$

bounded by the perpendicular bisectors of the segments $[c, c']$. Since $|c' - c| \ge 2$, the bisector is at distance at least 1 from c, so $c + B^n \subseteq V_c$. Cells of distinct centres have disjoint interiors and their union is $\mathbb{R}^n$. A packing is *saturated* if every point of $\mathbb{R}^n$ is within distance 2 of a centre; then every cell is contained in the ball of radius 2 about its centre.

Lemma 2.1 (Cells only shrink)**.** *If $\mathcal{C} \subseteq \mathcal{C}'$ are packings and $c \in \mathcal{C}$, then $V_c(\mathcal{C}') \subseteq V_c(\mathcal{C})$.*

Proof. $V_c(\mathcal{C}')$ is the intersection of the half-spaces $H_c(c')$ over the larger set of centres. □

Lemma 2.2 (Shell bound)**.** *Let $K \subseteq V_c(\mathcal{C})$ be a bounded set containing c with circumradius R about c, that is $|z - c| \le R$ for $z \in K$. If $|c' - c| \ge 2R$ then $K \subseteq H_c(c')$. In particular, if $V_c(\mathcal{C})$ has circumradius R about c, adding centres at distance at least $2R$ from c does not change V_c.*

Proof. For $z \in K$, $\langle z - c, c' - c \rangle \le |z - c|\,|c' - c| \le R\,|c' - c| \le \frac{1}{2}|c' - c|^2$. □

2.2. **Density from cells.** We recall the local argument in the form in which it would be used, to make clear what Theorem 1.1 destroys.

Proposition 2.3. *Suppose that $\omega > 0$ is such that every Voronoi cell of every saturated unit-ball packing of $\mathbb{R}^n$ has volume at least ω. Then $\Delta_n \le \mathrm{vol}(B^n)/\omega$.*

Proof. Every packing is contained in a saturated packing, and the density of the larger packing is at least that of the smaller, so it suffices to bound the density of a saturated packing $\mathcal{C}$. Its cells have circumradius at most 2 about their centres. For $R > 2$ the cells of the centres in the ball RB^n are disjoint up to measure zero and lie in $(R + 2)B^n$, so $|\mathcal{C} \cap RB^n|\,\omega \le \mathrm{vol}((R + 2)B^n)$, and the fraction of RB^n covered by the balls is at most $|\mathcal{C} \cap RB^n|\,\mathrm{vol}(B^n)/\,\mathrm{vol}(RB^n) \le \mathrm{vol}(B^n)(R + 2)^n/(\omega R^n) \to \mathrm{vol}(B^n)/\omega$. □

The proposition only asks for the bound on saturated packings, which is why Corollary 1.3 is stated for them: an example with a small cell in a non-saturated packing would not by itself close the door. The passage from Theorem 1.1 to saturated and periodic packings is Lemma 2.4 below.

Lemma 2.4 (Saturated periodic extensions)**.** *Every finite unit-ball packing $\mathcal{C}_0 \subset \mathbb{R}^n$ is contained in a periodic saturated unit-ball packing.*

Proof. Choose $L > 0$ so large that $\mathcal{C}_0$ lies in the open cube $(-L/2+2,\, L/2-2)^n$, and let $T = \mathbb{R}^n/L\mathbb{Z}^n$ be the torus. The image of $\mathcal{C}_0$ in T consists of points at pairwise distance at least 2 in the quotient metric, since the only translates that could come closer are those by non-zero vectors of $L\mathbb{Z}^n$, which move a point by at least L, and L exceeds the diameter of $\mathcal{C}_0$ by more than 4. A set of points of T at pairwise distance at least 2 has at most $\operatorname{vol}(T)/\operatorname{vol}(B^n)$ elements, so among such sets containing the image of $\mathcal{C}_0$ there is a maximal one, $\bar{\mathcal{C}}$. Its preimage $\mathcal{C}$ in $\mathbb{R}^n$ is a periodic packing containing $\mathcal{C}_0$. It is saturated: if a point $z \in \mathbb{R}^n$ were at distance at least 2 from all of $\mathcal{C}$, its image in T could be added to $\bar{\mathcal{C}}$. □

2.3. **Lattices, relevant vectors and holes.** Let $\Lambda \subset \mathbb{R}^n$ be a lattice whose minimal vectors have length 2, M the set of minimal vectors, and $P = V_0(\Lambda)$. The cells $P + \lambda$, $\lambda \in \Lambda$, tile $\mathbb{R}^n$, so $\operatorname{vol} P = \operatorname{covol}\Lambda$. A vector $m \in \Lambda \setminus \{0\}$ is *Voronoi relevant* if the bisector of $[0, m]$ contains a facet of P; Voronoi's criterion says that m is relevant if and only if $\pm m$ are the only shortest vectors in the coset $m + 2\Lambda$ (Voronoi [Vor08]; see Conway and Sloane [CS99, Ch. 21], Schürmann [Sch09] and [DSV09]). The vertices of P are the *holes* of Λ, the points of $\mathbb{R}^n$ that are local maxima of the distance to Λ; the *deep holes* are the holes at the largest distance, which is the covering radius ρ, and ρ is the circumradius of P about the origin.

We shall only use the following consequence of Voronoi's criterion, which is also the form in which it is verified in our computations.

Lemma 2.5 (Minimal vectors suffice)**.** *Let* $Q = \bigcap_{m\in M}\{z : \langle z, m\rangle \le \frac{1}{2}|m|^2\} = \bigcap_{m\in M}\{z : \langle z, m\rangle \le 2\}$. *Then* $P \subseteq Q$, *and if* $\operatorname{vol} Q = \operatorname{covol}\Lambda$ *then* $P = Q$; *in that case the relevant vectors of* Λ *are exactly the minimal vectors.*

Proof. P is the intersection of the half-spaces of all non-zero lattice vectors, Q of those of the minimal vectors, so $P \subseteq Q$. If the volumes agree, $Q \setminus P$ has measure zero; both are closed convex sets with non-empty interior, and a closed convex set is the closure of its interior, so $P = Q$. Each facet of Q lies on the bisector of a minimal vector, and each minimal vector is relevant for Q since otherwise it could be omitted from the intersection, which would change the volume. □

For the root lattices A_n, D_n, E_6, E_7, E_8 the determinants are $n+1$, 4, 3, 2, 1 in the usual normalisation with roots of squared length 2 (Conway and Sloane [CS99, Ch. 4]); scaling by $\sqrt{2}$ multiplies the covolume by $2^{n/2}$. For A_3, D_4 and the seven lattices in the table of Theorem 1.7 the computation of Section 10 finds $\operatorname{vol} Q = 2^{n/2}\sqrt{\det}$, so Lemma 2.5 applies and $P = Q$ is cut out by the minimal vectors. For E_7 this reads

$$\operatorname{vol} P = 2^{7/2}\sqrt{2} = 16, \tag{2.1}$$

and we derive $\det E_7 = 2$ in Section 4 from the unimodularity of E_8.

3. The one-point criterion

In this section $\Lambda \subset \mathbb{R}^n$ is a lattice with minimal distance 2, M is its set of minimal vectors, and we assume that the relevant vectors of Λ are exactly the minimal vectors, so that

$$P = V_0(\Lambda) = \bigcap_{m\in M}\{z :\ \langle z, m\rangle \le 2\}. \tag{3.1}$$

Then $\{0\} \cup M$ is a finite packing whose cell at the origin is P: the half-spaces that define $V_0(\{0\} \cup M)$ are exactly those in (3.1). We ask when one further centre can make this cell smaller.

Lemma 3.1 (Admissible radii)**.** *Let* u *be a unit vector and* $r > 0$. *The set* $\{0\} \cup M \cup \{ru\}$ *is a unit-ball packing if and only if*

$$r \ \ge\ r_{\min}(u) := \max\bigl(2,\ 2\,h_{\operatorname{conv} M}(u)\bigr), \qquad h_{\operatorname{conv} M}(u) = \max_{m\in M}\langle u, m\rangle. \tag{3.2}$$

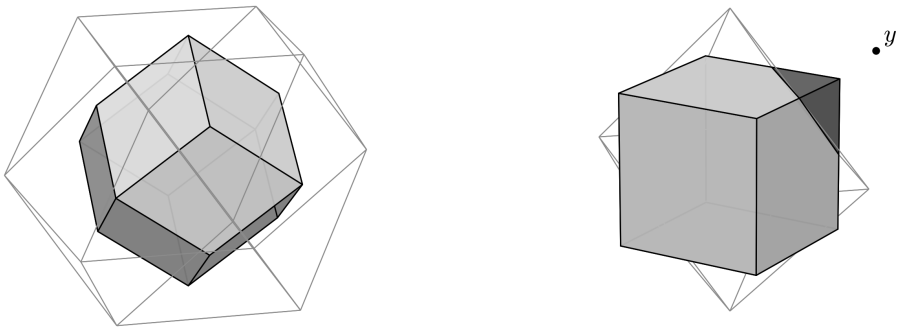


FIGURE 2. The criterion in three dimensions, with minimal distance 2. Left: the face-centred cubic lattice A_3; its cell, the rhombic dodecahedron (shaded), lies inside the convex hull of the twelve minimal vectors, a cuboctahedron (lines), and touches its boundary at the centres of the square faces, as $\rho = \sqrt{2}$ requires. No single extra centre can cut it. Right: the cubic lattice $\mathbb{Z}^3$, with $\rho = \sqrt{3}$; the corner of the cubic cell sticks out of the octahedron spanned by the six minimal vectors, and the extra centre $y = \frac{4}{3}(1,1,1)$, which touches the three balls at $2e_1, 2e_2, 2e_3$, cuts the dark corner off the cell.

Proof. The new centre must be at distance at least 2 from the origin, which is $r \geq 2$, and from each $m \in M$. Since $|m| = 2$,

$$|ru - m|^2 = r^2 - 2r\langle u, m\rangle + 4 \;\geq\; 4 \quad\Longleftrightarrow\quad r \;\geq\; 2\langle u, m\rangle,$$

and the maximum of $\langle u, m\rangle$ over the finite set M is the support function of its convex hull at u. □

Lemma 3.2 (The cut)**.** *For $y = ru$ the cell of the origin in $\{0\} \cup M \cup \{y\}$ is $P \cap \{z : \langle z, u\rangle \leq r/2\}$, and its volume is smaller than* $\mathrm{vol}\, P$ *if and only if $h_P(u) > r/2$.*

Proof. The bisector of $[0, y]$ is $\{z : \langle z, y\rangle = \frac{1}{2}|y|^2\} = \{z : \langle z, u\rangle = r/2\}$, which gives the description of the cell. If $h_P(u) \leq r/2$ the half-space contains P and nothing is removed. If $h_P(u) > r/2$, let $z_0 \in P$ with $\langle z_0, u\rangle = h_P(u)$; every neighbourhood of z_0 meets the interior of P (a convex body is the closure of its interior), and the points of the interior close to z_0 satisfy $\langle z, u\rangle > r/2$, so a set of positive measure is removed. □

Lemma 3.3 (Facets of $\mathrm{conv}(M)$)**.** $\mathrm{conv}(M) = \{x \in \mathbb{R}^n : \; \langle x, v\rangle \leq 2$ *for every vertex v of P*$\}$.

Proof. By (3.1), $P = \{z : \; \langle z, m/2\rangle \leq 1 \text{ for all } m \in M\}$, so P is the polar body of $\frac{1}{2}\,\mathrm{conv}(M)$; this is the same duality as between a reflexive polytope and its dual in toric geometry (see [BSN26]). Both are compact convex sets with the origin in the interior, so by the bipolar theorem $\frac{1}{2}\,\mathrm{conv}(M) = P^\circ = \{x : \; \langle x, z\rangle \leq 1 \text{ for } z \in P\}$, and it is enough to test z at the vertices of P. Multiplying by 2 gives the claim. □

Proof of Theorem 1.6. By Lemmas 3.1 and 3.2, a point y on the ray through u that shrinks the cell exists if and only if

$$h_P(u) \;>\; \tfrac{1}{2} r_{\min}(u) = \max(1,\; h_{\mathrm{conv}\, M}(u)), \tag{3.3}$$

and then every r with $r_{\min}(u) \leq r < 2h_P(u)$ works. We claim that (3.3) is equivalent to $h_P(u) > h_{\mathrm{conv}\, M}(u)$. One direction is trivial. For the other, suppose $h_P(u) > h_{\mathrm{conv}\, M}(u)$ and $h_{\mathrm{conv}\, M}(u) < 1$. The unit ball lies in P, so $h_P(u) \geq 1$, and $h_P(u) = 1$ would mean that the hyperplane $\{\langle z, u\rangle = 1\}$ supports P at the point u of the unit sphere, so that u lies on the boundary of P, hence on a facet $\{\langle z, m\rangle = 2\}$ of P, hence $\langle u, m\rangle = 2$ and $u = m/2$; but then

$h_{\operatorname{conv} M}(u) \geq \langle m/2, m\rangle = 2$, against $h_{\operatorname{conv} M}(u) < 1$. So $h_P(u) > 1$, and (3.3) holds. This proves the first sentence of the theorem.

The support function of a compact convex set determines it, and $h_K \leq h_{K'}$ pointwise if and only if $K \subseteq K'$. So no direction admits a shrinking centre exactly when $P \subseteq \operatorname{conv} M$.

Let v be a deep hole, $\rho = |v|$, $u = v/\rho$. Since $v \in P$, $\langle v, m\rangle \leq 2$ for all $m \in M$, that is $\langle u, m\rangle \leq 2/\rho$; the facets of P through v give equality, so

$$h_{\operatorname{conv} M}(u) = \frac{2}{\rho}, \qquad h_P(u) = \rho, \tag{3.4}$$

the second because ρ is the circumradius of P about the origin and is attained at v. If $P \subseteq \operatorname{conv} M$ then $\rho \leq 2/\rho$, so $\rho \leq \sqrt{2}$. Conversely, if $\rho \leq \sqrt{2}$, then for $z \in P$ and every vertex v' of P we have $\langle z, v'\rangle \leq |z|\,|v'| \leq \rho^2 \leq 2$, so $z \in \operatorname{conv} M$ by Lemma 3.3. If $\rho > \sqrt{2}$ then $h_P(u) = \rho > 2/\rho = h_{\operatorname{conv} M}(u)$ and a shrinking centre exists on the ray; by (3.2) the smallest admissible radius is $r_{\min}(u) = \max(2, 4/\rho)$, and the corresponding bisector is at distance $\max(1, 2/\rho) < \rho$ from the origin, so it cuts P near v. □

Remark 3.4. The quantity $\rho/(\text{packing radius})$ is the ratio of covering radius to packing radius, which is at least 1 and equals $\sqrt{2}$ for A_3, D_4, E_8 and the Leech lattice. Theorem 1.6 explains, in retrospect, why the lattices in the solved dimensions and the lattice of Musin's conjecture are the ones for which the local approach is not excluded at the first step: for every other root lattice of dimension at least 4 a single extra ball, placed in a deep hole direction as close to the origin as the packing condition allows, makes a Voronoi cell smaller than the lattice cell.

Remark 3.5. For a lattice that fails the criterion, the deep-hole direction need not be the direction in which the most volume can be removed, and in Section 4 we remove far more by using many directions at once. The criterion is a statement about one extra centre; it says nothing about packings in which the lattice neighbourhood itself is distorted.

4. The lattice E_7 and its deep holes

4.1. **Coordinates.** We use the standard model of E_8 in $\mathbb{R}^8$,

$$E_8 = \Big\{x \in \mathbb{Z}^8 \cup (\mathbb{Z} + \tfrac{1}{2})^8 : \ \textstyle\sum_i x_i \equiv 0 \pmod 2\Big\},$$

an even unimodular lattice whose 240 roots (vectors of squared length 2) are the 112 vectors $\pm e_i \pm e_j$ $(i < j)$ and the 128 vectors $(\pm\frac{1}{2}, \dots, \pm\frac{1}{2})$ with an even number of minus signs [CS99, Ch. 4, §8.1]. Let $a = e_1 + e_2$, a root, and

$$L = \{x \in E_8 : \ \langle x, a\rangle = 0\} = \{x \in E_8 : \ x_1 + x_2 = 0\}.$$

The orthogonal complement of a root in E_8 is a copy of E_7 [CS99, Ch. 4, §8.3]; everything we use about L is derived below from the coordinates, so the reader may take L as the definition of E_7.

Lemma 4.1 (Roots of L)**.** *L contains exactly* 126 *roots: the* 60 *vectors $\pm e_i \pm e_j$ with $3 \leq i < j \leq 8$, the two vectors $\pm(e_1 - e_2)$, and the* 64 *vectors $(\pm\frac{1}{2}, \mp\frac{1}{2}, x_3, \dots, x_8)$ with $x_i = \pm\frac{1}{2}$ and an odd number of minus signs among $x_3, \dots, x_8$.*

Proof. A root $\pm e_i \pm e_j$ has $x_1 + x_2 = 0$ exactly when $\{i, j\} \cap \{1, 2\} = \emptyset$ (60 vectors) or $\{i, j\} = \{1, 2\}$ with opposite signs (2 vectors). A half-integral root has $x_1 + x_2 = 0$ exactly when $x_1 = -x_2$; this uses one minus sign among the first two coordinates, so the remaining six must carry an odd number of minus signs to make the total even: 2 choices for (x_1, x_2) and $2^5 = 32$ for the rest. □

Lemma 4.2 (Determinant)**.** $\det L = 2$. *Consequently the lattice $\Lambda = \sqrt{2}\,L$, whose minimal vectors have length* 2*, has covolume* $\operatorname{vol} P = 2^{7/2}\sqrt{2} = 16$.

Proof. The map $x \mapsto \langle x, a\rangle = x_1 + x_2$ sends E_8 onto $\mathbb{Z}$ (it takes the value 1 at $(\frac{1}{2}, \dots, \frac{1}{2})$ and the value 0 at $e_1 - e_2$). Its kernel is L. The sublattice $\mathbb{Z}a \oplus L$ consists of the $x \in E_8$ with $\langle x, a\rangle$ even, since the component of x along a is $\frac{1}{2}\langle x, a\rangle\, a$ and lies in $\mathbb{Z}a$ exactly when $\langle x, a\rangle$ is even. So $\mathbb{Z}a \oplus L$ has index 2 in E_8, and

$$\det(\mathbb{Z}a \oplus L) = \det(\mathbb{Z}a)\det L = 2\det L = 2^2 \det E_8 = 4.$$

Hence $\det L = 2$, $\operatorname{covol} L = \sqrt{2}$, and scaling by $\sqrt{2}$ multiplies 7-dimensional volumes by $2^{7/2}$. □

From now on $\Lambda = \sqrt{2}\,L$, $M = \sqrt{2}\cdot\{\text{roots of } L\}$ is its set of 126 minimal vectors, $u_m = m/2$ is the unit vector along $m \in M$, and

$$P = \bigcap_{m \in M} \{z : \ \langle z, u_m\rangle \le 1\} \tag{4.1}$$

is the polytope cut out by the minimal vectors. By Lemma 2.5 and the exact computation $\operatorname{vol} P = 16$ of Section 10, P is the Voronoi cell of Λ and the relevant vectors of Λ are its minimal vectors. It is convenient to switch between Λ and L: a point z of the Λ-picture corresponds to $w = z/\sqrt{2}$ in the L-picture, where P becomes $P_L = \{w : \langle w, \alpha\rangle \le 1 \text{ for all roots } \alpha\}$; inner products of the form $\langle z, u_m\rangle$ become $\langle w, \alpha\rangle$ with $\alpha = m/\sqrt{2}$ the root.

4.2. **A deep hole and the twenty-seven roots through it.** Let

$$v_L = -\tfrac{1}{2}(0,0,1,1,1,1,1,1), \qquad v = \sqrt{2}\,v_L, \qquad u = \frac{v}{|v|} = -\frac{1}{\sqrt{6}}(0,0,1,1,1,1,1,1). \tag{4.2}$$

Lemma 4.3. *$|v_L|^2 = \frac{3}{2}$, so $|v| = \sqrt{3}$. For every root α of L, $\langle v_L, \alpha\rangle \in \{1, 0, -1\}$, and the value 1 is taken by exactly 27 roots: the 15 roots $-e_i - e_j$ with $3 \le i < j \le 8$, and the 12 roots $(\pm\frac{1}{2}, \mp\frac{1}{2}, x_3, \dots, x_8)$ in which exactly one of $x_3, \dots, x_8$ equals $+\frac{1}{2}$. Consequently $v_L \in P_L$, and v_L lies on exactly 27 facets of P_L.*

Proof. $\langle v_L, \alpha\rangle = -\frac{1}{2}(\alpha_3 + \cdots + \alpha_8)$. For $\alpha = \pm e_i \pm e_j$ with $i, j \ge 3$ the sum $\alpha_3 + \cdots + \alpha_8$ is 2, 0 or -2, giving -1, 0 or 1, with 1 exactly for $-e_i - e_j$; for $\alpha = \pm(e_1 - e_2)$ it is 0. For a half-integral root the six coordinates $\alpha_3, \dots, \alpha_8$ carry an odd number $k \in \{1, 3, 5\}$ of minus signs (Lemma 4.1), so their sum is $\frac{1}{2}(6 - 2k) \in \{2, 0, -2\}$ and $\langle v_L, \alpha\rangle \in \{-1, 0, 1\}$, with 1 exactly when $k = 5$, that is when exactly one of the six coordinates is $+\frac{1}{2}$: 6 positions times 2 choices of (α_1, α_2). □

Write S for this set of 27 roots and $u_\alpha = \alpha/\sqrt{2}$ for the corresponding unit normals of the facets of P. In the Λ-picture Lemma 4.3 says $\langle v, u_\alpha\rangle = 1$ for $\alpha \in S$, $\langle v, u_\beta\rangle \in \{0, -1\}$ for $\beta \notin S$, and $\langle u, u_\alpha\rangle = 1/\sqrt{3}$ for $\alpha \in S$.

Lemma 4.4 (Gram matrix of S)**.** *For distinct $\alpha, \beta \in S$ the inner product $\langle \alpha, \beta\rangle$ is 1 or 0; each $\alpha \in S$ has 16 partners with $\langle \alpha, \beta\rangle = 1$ and 10 with $\langle \alpha, \beta\rangle = 0$.*

Proof. Call the roots $-e_i - e_j$ of type I and the half-integral ones of type II, and for a type II root let $p \in \{3, \dots, 8\}$ be the position of its $+\frac{1}{2}$ and $\varepsilon = \pm 1$ the sign of its first coordinate. Two type I roots $-e_i - e_j$, $-e_k - e_l$ have inner product $|\{i, j\} \cap \{k, l\}| \in \{0, 1\}$; for fixed $\{i, j\}$ there are $2 \cdot 4 = 8$ pairs $\{k, l\}$ meeting it in one element and $\binom{4}{2} = 6$ disjoint from it. A type I root $-e_i - e_j$ and a type II root have inner product $-(x_i + x_j)$, which is 1 if $p \notin \{i, j\}$ and 0 if $p \in \{i, j\}$; for fixed $\{i, j\}$ this gives $4 \cdot 2 = 8$ type II roots with value 1 and $2 \cdot 2 = 4$ with value 0, and for a fixed type II root $\binom{5}{2} = 10$ type I roots with value 1 and 5 with value 0. Two type II roots with data (p, ε), (p', ε') have inner product $\frac{1}{2}\varepsilon\varepsilon' + \frac{1}{4}\sum_{i \ge 3} \sigma_i \sigma_i'$, where $\sigma_i = \pm 1$ are the signs of the last six coordinates; the last sum is $\frac{6}{4}$ if $p = p'$ and $\frac{4-2}{4} = \frac{1}{2}$ if $p \ne p'$. So the inner product is 1 when ($\varepsilon = \varepsilon'$, $p \ne p'$: 5 roots) or ($\varepsilon \ne \varepsilon'$, $p = p'$: 1 root), and 0 when ($\varepsilon \ne \varepsilon'$, $p \ne p'$: 5 roots). Adding up, a type I root has $8 + 8 = 16$ partners with value 1 and $6 + 4 = 10$ with value 0, and a type II root has $10 + 6 = 16$ and $5 + 5 = 10$. □

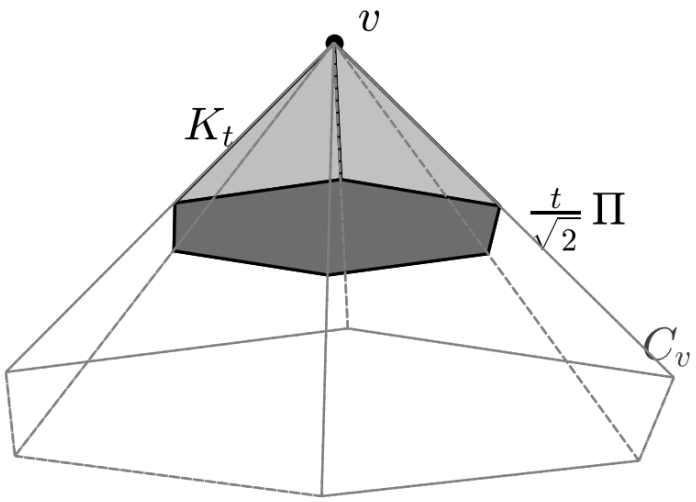


FIGURE 3. The cone C_v at the deep hole v, its truncation at height t along $-u$ (the cap K_t, shaded), and the cross-section, which is a scaled copy of the polar Π of the Schläfli polytope 2_{21}. The figure is a three-dimensional schematic of a seven-dimensional object; the hexagonal pyramid stands for the cone over 27 facets.

Lemma 4.5 (Vertex)**.** *v is a vertex of P, and $|v| = \sqrt{3}$ is the circumradius of P: v is a deep hole of Λ and $\rho = \sqrt{3}$.*

Proof. The 27 roots of S span the hyperplane $a^\perp$: the type I roots span the coordinate space of $e_3, \dots, e_8$, and any type II root adds the direction $e_1 - e_2$. So the 27 facet hyperplanes through v meet only in v, which is therefore a vertex. That no vertex of P is farther from the origin than $\sqrt{3}$ is part of the vertex enumeration of Section 10: the 632 vertices of P have squared norm 3 (56 of them) or $\frac{7}{4}$ (576 of them). This agrees with the known covering radius $\sqrt{3/2}$ of E_7 in the root normalisation [CS99, Ch. 4, §8.3]. □

The 56 deep holes, in the L-picture, are the vectors

$$\begin{aligned} &\tfrac{1}{2}(0,0,\varepsilon_3,\dots,\varepsilon_8),\ \varepsilon_i = \pm 1 \text{ with an even number of } + \text{ signs} \quad (32 \text{ vectors}),\\ &(\pm\tfrac{1}{2}, \mp\tfrac{1}{2}, \pm e_i),\ 3 \le i \le 8 \quad (24 \text{ vectors}), \end{aligned} \tag{4.3}$$

and all have squared norm $\frac{3}{2}$; the list is read off the vertex enumeration, and the pairwise inner products are computed by hand in Lemma 4.11 below.

4.3. **The cone at the deep hole and its cross-section.** Let

$$C_v = \{s \in \mathbb{R}^7 :\ \langle s, u_\alpha\rangle \le 0 \text{ for all } \alpha \in S\}$$

be the tangent cone of P at v generated by the 27 facets through v, and for $t > 0$ let

$$K_t = P \cap \{z :\ \langle z, u\rangle \ge \sqrt{3} - t\} \tag{4.4}$$

be the cap of P of height t at v. The plan is to show that for $t \le 1/\sqrt{3}$ the cap is a truncation of the cone, to identify the cross-section of the cone, and to read off the volume of the cap from a single exact computation.

Lemma 4.6 (Cross-section)**.** *Write $s = -t'u + x$ with $x \perp u$ and $t' \ge 0$. Then $s \in C_v$ if and only if $x \in \frac{t'}{\sqrt{2}}\,\Pi$, where*

$$\Pi = \{x \in u^\perp :\ \langle x, n_\alpha\rangle \le 1 \text{ for all } \alpha \in S\}, \qquad n_\alpha = \sqrt{\tfrac{3}{2}}\,(u_\alpha - \langle u_\alpha, u\rangle\, u),$$

and the n_α are 27 unit vectors of $u^\perp$ with $\langle n_\alpha, n_\beta\rangle = \frac{1}{4}$ for $\langle \alpha,\beta\rangle = 1$ and $\langle n_\alpha, n_\beta\rangle = -\frac{1}{2}$ for $\langle \alpha,\beta\rangle = 0$. Consequently $\operatorname{conv}\{n_\alpha : \alpha \in S\}$ is congruent to the Schläfli polytope 2_{21} inscribed in the unit sphere of $\mathbb{R}^6$, and Π is its polar.

Proof. For $\alpha \in S$, $\langle u_\alpha, u\rangle = 1/\sqrt{3}$, so $\langle s, u_\alpha\rangle = -\frac{t'}{\sqrt{3}} + \langle x, u_\alpha - \langle u_\alpha, u\rangle\, u\rangle$, and the projected vector $u_\alpha - \langle u_\alpha, u\rangle\, u$ has squared length $1 - \frac{1}{3} = \frac{2}{3}$. Dividing by $\sqrt{2/3}$ turns the condition $\langle s, u_\alpha\rangle \le 0$ into $\langle x, n_\alpha\rangle \le \frac{t'}{\sqrt{3}}\sqrt{\frac{3}{2}} = \frac{t'}{\sqrt{2}}$. The inner products of the n_α follow from Lemma 4.4: $\langle u_\alpha, u_\beta\rangle = \frac{1}{2}\langle \alpha,\beta\rangle \in \{\frac{1}{2}, 0\}$ and

$$\langle n_\alpha, n_\beta\rangle = \tfrac{3}{2}\big(\langle u_\alpha, u_\beta\rangle - \tfrac{1}{3}\big) \in \{\tfrac{1}{4},\ -\tfrac{1}{2}\}.$$

A set of 27 unit vectors in $\mathbb{R}^6$ in which each vector has 16 partners at inner product $\frac{1}{4}$ and 10 at $-\frac{1}{2}$, with the incidence pattern of Lemma 4.4, is determined up to isometry by its Gram matrix, and this is the Gram matrix of the vertices of 2_{21}, the 27 vertices of which correspond to the lines on a cubic surface, two lines being skew or meeting according as the inner product is $\frac{1}{4}$ or $-\frac{1}{2}$ (Coxeter [Cox40]).[3] □

Lemma 4.7 (Circumradius of Π)**.** *The polar Π has 99 vertices, 27 of them at distance 2 from the origin and 72 at distance $\sqrt{8/3}$; its circumradius is 2.*

Proof. The vertices of Π are the points x with $\langle x, n_\alpha\rangle = 1$ for the vertices n_α of a facet of 2_{21}, that is $x = c/|c|^2$ where c is the foot of the perpendicular from the origin to the facet hyperplane; so the distances of the vertices of Π are the reciprocals of the distances of the facets of 2_{21}. The facets of 2_{21} are 27 regular 5-dimensional cross-polytopes and 72 regular 5-simplices (Coxeter [Cox40]), and this is confirmed in Section 10 by computing the facets of $\operatorname{conv}\{n_\alpha\}$: 27 facets with 10 vertices and 72 with 6. A cross-polytope facet contains, together with each of its vertices n, the antipodal vertex n' of the cross-polytope, with $\langle n, n'\rangle = -\frac{1}{2}$, so its centre $c = \frac{1}{2}(n + n')$ has $|c|^2 = \frac{1}{4}(1 + 1 - 1) = \frac{1}{4}$ and the facet is at distance $\frac{1}{2}$. A simplex facet has six vertices with pairwise inner products $\frac{1}{4}$, centre $c = \frac{1}{6}\sum n$, $|c|^2 = \frac{1}{36}(6 + 30\cdot\frac{1}{4}) = \frac{3}{8}$, and is at distance $\sqrt{3/8}$. The reciprocals are 2 and $\sqrt{8/3}$. □

Lemma 4.8 (The cap is a truncated cone)**.** *For $0 < t \le 1/\sqrt{3}$,*

$$K_t = v + \big(C_v \cap \{s :\ \langle s, -u\rangle \le t\}\big) = v + t\,\big(C_v \cap \{s :\ \langle s, -u\rangle \le 1\}\big),$$

and $\operatorname{vol} K_t = t^7\,\kappa$ with $\kappa = \operatorname{vol}\big(C_v \cap \{\langle s, -u\rangle \le 1\}\big)$.

Proof. The second equality is the homothety $s \mapsto ts$, which preserves the cone and scales the slab, and the volume formula follows. For the first equality, if $z \in K_t$ then $z \in P$ gives $\langle z, u_\alpha\rangle \le 1 = \langle v, u_\alpha\rangle$ for $\alpha \in S$, so $s = z - v \in C_v$, and $\langle z, u\rangle \ge \sqrt{3} - t$ gives $\langle s, -u\rangle \le t$. Conversely let $s = -t'u + x \in C_v$ with $0 \le t' \le t$ and $x \perp u$, and $z = v + s$. We must show $\langle z, u_\beta\rangle \le 1$ for every root $\beta \notin S$ (for $\alpha \in S$ it holds by the definition of C_v, and $\langle z, u\rangle = \sqrt{3} - t' \ge \sqrt{3} - t$). By Lemma 4.3, $\langle v, u_\beta\rangle \in \{0, -1\}$, so $\langle u, u_\beta\rangle \in \{0, -1/\sqrt{3}\}$ and

$$\langle z, u_\beta\rangle = \langle v, u_\beta\rangle - t'\langle u, u_\beta\rangle + \langle x, u_\beta\rangle \le \langle v, u_\beta\rangle + \frac{t'}{\sqrt{3}}\,\mathbf{1}[\langle v, u_\beta\rangle = -1] + |x|.$$

If $\langle v, u_\beta\rangle = -1$ the right-hand side is at most $-1 + \frac{1}{3} + |x|$, and if $\langle v, u_\beta\rangle = 0$ it is $|x|$; in both cases it suffices to have $|x| \le 1$. By Lemmas 4.6 and 4.7, $x \in \frac{t'}{\sqrt{2}}\Pi$ has $|x| \le \frac{t'}{\sqrt{2}}\cdot 2 = \sqrt{2}\,t' \le \sqrt{2}/\sqrt{3} < 1$. □

The bound in the proof has room to spare: the exact computation shows that $\max\langle z, u_\beta\rangle$ over the vertices of $K_{1/\sqrt{3}}$ and all $\beta \notin S$ equals $\frac{2}{3}$.

[3] In the language of root systems the n_α are the 27 weights of the minuscule representation of E_6, rescaled to unit length; their inner products $\frac{4}{3}$, $\frac{1}{3}$, $-\frac{2}{3}$ become 1, $\frac{1}{4}$, $-\frac{1}{2}$. The identification with 2_{21} is only used to name the cross-section; every number we need is computed directly from the coordinates.

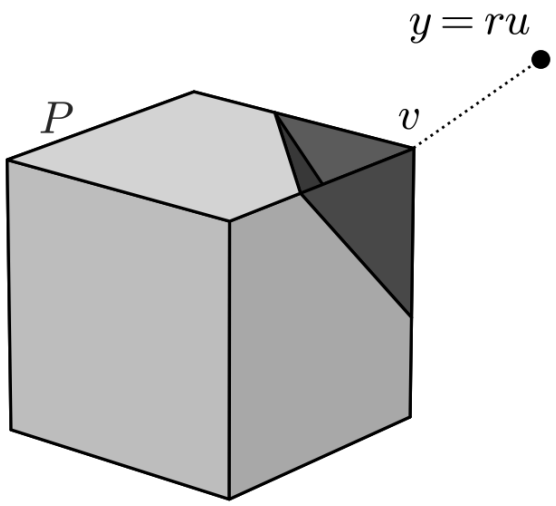


FIGURE 4. The construction behind Theorem 1.1: the Voronoi cell P of the origin, the deep hole v at distance $\rho = \sqrt{3}$ in the direction u, and the extra centre $y = ru$ beyond it. The bisector $\langle z, u\rangle = r/2$ cuts the dark cap, of height $\sqrt{3} - r/2$, off P. Three-dimensional schematic of a seven-dimensional picture.

4.4. **The volume of the cap.**

Proposition 4.9. $\operatorname{vol} K_{1/\sqrt{3}} = \dfrac{1}{105}$. *Hence* $\kappa = 3^{7/2}/105$ *and, for* $0 < t \le 1/\sqrt{3}$,

$$\operatorname{vol} K_t = \frac{(\sqrt{3}\,t)^7}{105}. \tag{4.5}$$

Proof. The first statement is a finite computation in exact rational arithmetic, carried out in two independent ways in Section 10. In the first, the polytope $P \cap \{\langle z, u\rangle \le 2/\sqrt{3}\}$, which has 127 facets and 730 vertices, is enumerated and its volume comes out as $\frac{1679}{105}$; subtracting from 16 gives $\frac{1}{105}$. In the second, the cross-section $P \cap \{\langle z, u\rangle = 2/\sqrt{3}\}$, a 6-polytope with 27 facets and 99 vertices, is enumerated; by Lemma 4.8 the cap is the convex hull of the deep hole v and these 99 points, and the volume of that convex hull is computed directly as $\frac{1}{105}$. The rest is Lemma 4.8: $\kappa = \operatorname{vol} K_{1/\sqrt{3}} \cdot (\sqrt{3})^7$. □

Corollary 4.10 (Volume of the polar of 2_{21})**.** *The polar* Π *of the Schläfli polytope* 2_{21} *inscribed in the unit sphere of* $\mathbb{R}^6$ *has volume* $\operatorname{vol}_6 \Pi = \dfrac{72\sqrt{3}}{5}$.

Proof. By Lemma 4.6 the cross-section of the cone at height t is $\frac{t}{\sqrt{2}}\Pi$, so

$$\operatorname{vol} K_t = \tfrac{1}{7}\, t\, \big(\tfrac{t}{\sqrt{2}}\big)^6 \operatorname{vol}_6 \Pi = \frac{t^7}{56} \operatorname{vol}_6 \Pi.$$

Comparing with (4.5), $\operatorname{vol}_6 \Pi = \frac{56 \cdot 3^{7/2}}{105} = \frac{8 \cdot 27\sqrt{3}}{15} = \frac{72\sqrt{3}}{5}$. □

4.5. **Proof of Theorem 1.1.** Let $\frac{4}{\sqrt{3}} \le r \le 2\sqrt{3}$ and $y_r = ru$; Figure 4 shows the configuration.

Packing. By Lemma 3.1 we need $r \ge \max(2, 2h_{\operatorname{conv} M}(u))$. By (3.4) with $\rho = \sqrt{3}$, $h_{\operatorname{conv} M}(u) = 2/\sqrt{3}$, so the condition is $r \ge 4/\sqrt{3}$, which holds; and $4/\sqrt{3} > 2$. The centres $m \in M$ at distance exactly 2 from $y_{4/\sqrt{3}}$ are those with $\langle u, m\rangle = 2/\sqrt{3}$, that is $\langle v, u_m\rangle = 1$: the 27 minimal vectors $\sqrt{2}\alpha$, $\alpha \in S$.

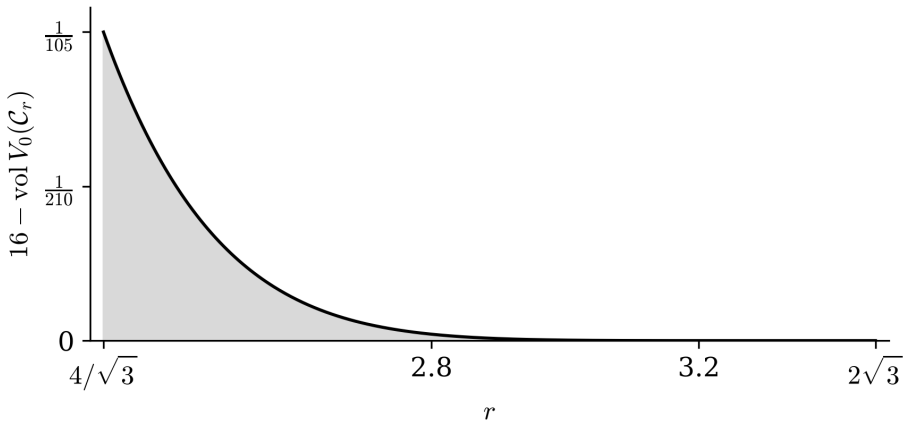


FIGURE 5. The volume removed from the cell of the origin by one extra centre at distance r in a deep-hole direction, as a function of r. Below $r = 4/\sqrt{3}$ the extra ball would overlap one of the 27 balls around the hole; at $r = 2\sqrt{3}$ the bisector passes through the hole and nothing is removed.

The cell. By Lemma 3.2 the cell of the origin is $P \cap \{\langle z, u\rangle \le r/2\}$, and $r/2 \le \sqrt{3} = h_P(u)$. What is removed is the cap K_t with $t = \sqrt{3} - r/2 \in [0, 1/\sqrt{3}]$ (up to the hyperplane, of measure zero), so by Proposition 4.9

$$\operatorname{vol} V_0(\mathcal{C}_r) = 16 - \frac{(\sqrt{3}t)^7}{105} = 16 - \frac{1}{105}\Big(3 - \frac{\sqrt{3}}{2}r\Big)^7.$$

At $r = 4/\sqrt{3}$ the bracket is $3 - 2 = 1$.

Other packings. The cell $P \cap \{\langle z, u\rangle \le r/2\}$ lies in P, whose circumradius about the origin is $\sqrt{3}$. By Lemma 2.2, centres at distance at least $2\sqrt{3}$ from the origin do not change it. By Lemma 2.1, any packing containing $\mathcal{C}_r$ has a cell at the origin contained in $P \cap \{\langle z, u\rangle \le r/2\}$, hence of volume at most the value above. □

4.6. **All fifty-six deep holes.**

Lemma 4.11 (Inner products of the deep holes)**.** *Let $v_1, \dots, v_{56}$ be the deep holes of* (4.3)*, in the L-picture, and $u_i = v_i/|v_i|$. For $i \ne j$, $\langle u_i, u_j\rangle \in \{\frac{1}{3}, -\frac{1}{3}, -1\}$.*

Proof. All v_i have squared norm $\frac{3}{2}$, so $\langle u_i, u_j\rangle = \frac{2}{3}\langle v_i, v_j\rangle$ and we show $\langle v_i, v_j\rangle \in \{\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}\}$ for $i \ne j$. Two holes of the first kind, $\frac{1}{2}(0, 0, \varepsilon)$ and $\frac{1}{2}(0, 0, \varepsilon')$, have inner product $\frac{1}{4}\sum \varepsilon_i\varepsilon'_i = \frac{1}{4}(6 - 2k)$ where k is the number of positions where they differ; k is even because both sign vectors have an even number of $+$ signs, and $k \ne 0$, so $k \in \{2, 4, 6\}$ and the inner product is $\frac{1}{2}$, $-\frac{1}{2}$ or $-\frac{3}{2}$. A hole of the first kind and one of the second, $(\pm\frac{1}{2}, \mp\frac{1}{2}, \pm e_i)$, have inner product $\pm\frac{1}{2}\varepsilon_i = \pm\frac{1}{2}$. Two holes of the second kind, $(\varepsilon\frac{1}{2}, -\varepsilon\frac{1}{2}, \sigma e_i)$ and $(\varepsilon'\frac{1}{2}, -\varepsilon'\frac{1}{2}, \sigma' e_j)$, have inner product $\frac{1}{2}\varepsilon\varepsilon' + \sigma\sigma'[i = j]$, which for distinct holes takes the values $\frac{1}{2}$ ($\varepsilon = \varepsilon'$, $i \ne j$), $-\frac{1}{2}$ ($\varepsilon = \varepsilon'$, $i = j$, $\sigma \ne \sigma'$), $-\frac{1}{2}$ ($\varepsilon \ne \varepsilon'$, $i \ne j$), $\frac{1}{2}$ ($\varepsilon \ne \varepsilon'$, $i = j$, $\sigma = \sigma'$) and $-\frac{3}{2}$ ($\varepsilon \ne \varepsilon'$, $i = j$, $\sigma \ne \sigma'$). □

Proof of Theorem 1.2. Put $r = 4/\sqrt{3}$ and $y_i = r\,u_i$ in the Λ-picture (the directions u_i are the same in both pictures). Each y_i is at distance $r > 2$ from the origin and at distance at least 2 from every minimal vector, by the packing step of the proof of Theorem 1.1 applied to the deep hole v_i; the argument only used $|v_i| = \sqrt{3}$ and $\langle v_i, u_m\rangle \le 1$, which hold for every deep hole. For $i \ne j$, by Lemma 4.11,

$$|y_i - y_j|^2 = 2r^2(1 - \langle u_i, u_j\rangle) = \frac{32}{3}(1 - \langle u_i, u_j\rangle) \;\ge\; \frac{32}{3}\cdot\frac{2}{3} = \frac{64}{9},$$

so $|y_i - y_j| \ge \frac{8}{3} > 2$. Hence the 183 points form a packing. The cell of the origin is $P \cap \bigcap_i \{\langle z, u_i\rangle \le 2/\sqrt{3}\}$, and its volume is the exact computation of Section 10: the polytope

has 182 facets and 5364 vertices, and its volume is $\frac{29}{30}\cdot 16 = \frac{232}{15}$. Equivalently, the 56 caps $K^{(i)}_{1/\sqrt{3}}$ at the deep holes have pairwise intersections of measure zero and are removed together, $56\cdot\frac{1}{105} = \frac{8}{15}$. □

Proof of Corollary 1.3. Apply Lemma 2.4 to the finite packing of Theorem 1.2 and then Lemma 2.1: the cell of the origin in the periodic saturated extension lies in the cell computed in Theorem 1.2, so its volume is at most $232/15 < 16$. If a bound $\operatorname{vol} V_c \geq \omega$ held for all cells of all saturated packings, then $\omega \leq 232/15$ and Proposition 2.3 would give only $\Delta_7 \leq \operatorname{vol}(B^7)\cdot 15/232$, which exceeds $\pi^3/105$. □

Remark 4.12. The extra balls pay for the small cell. In the finite packing the cell of each y_i is unbounded; in a saturated extension it is bounded by the balls that saturation adds, and the volume lost at the origin reappears in the cells of the y_i and of their new neighbours. This is exactly the situation that Hales's proof in dimension 3 handles by averaging with weights over a neighbourhood of each centre rather than by bounding each cell [Hal05, Hal12]. Nothing in this paper says anything against the possibility of such a weighted argument in dimension 7; what it says is that the unweighted one is unavailable.

5. The smallest cell around an E_7 shell

In this section $\mathcal{C}$ is a unit-ball packing of $\mathbb{R}^7$ that contains $\{0\}\cup M$, the origin together with the 126 minimal vectors of $\Lambda = \sqrt{2}\,E_7$. Nothing is assumed about the other centres. The cell of the origin is

$$V_0(\mathcal{C}) = P \cap \bigcap_{y\in\mathcal{C}\setminus(\{0\}\cup M)} H(y), \qquad H(y) = \{z :\ \langle z,y\rangle \leq \tfrac{1}{2}|y|^2\},$$

because the half-spaces of the minimal vectors already cut out P (Lemma 2.5 and $\operatorname{vol} P = 16$, Lemma 4.2).

Lemma 5.1 (The shell inequality)**.** *For every $y \in \mathcal{C}\setminus(\{0\}\cup M)$ and every $z \in \operatorname{conv}(M)$,*

$$\langle z,y\rangle \leq \tfrac{1}{2}|y|^2.$$

In other words $\operatorname{conv}(M) \subseteq H(y)$.

Proof. For $m \in M$ we have $|m| = 2$ and, since y and m are distinct centres, $|y-m| \geq 2$. Hence

$$\langle m,y\rangle = \tfrac{1}{2}(|m|^2 + |y|^2 - |y-m|^2) \leq \tfrac{1}{2}(4 + |y|^2 - 4) = \tfrac{1}{2}|y|^2.$$

The left-hand side is linear in m, so the inequality persists for convex combinations of points of M. □

The lemma is false for $y \in M$ itself, where $\langle y,y\rangle = 4 > 2$; this is why the cell is P intersected with the other half-spaces, and not $\operatorname{conv}(M)$ intersected with all of them.

By Lemma 3.3, $\operatorname{conv}(M)$ is cut out by the inequalities $\langle x,v\rangle \leq 2$ over the vertices v of P; for a deep hole $v = \sqrt{3}\,u$ the inequality reads $\langle x,u\rangle \leq 2/\sqrt{3}$. So the facets of $\operatorname{conv}(M)$ come in two kinds, 56 facets dual to the deep holes and 576 dual to the shallow holes, and the 56 facets of the first kind are exactly the cuts of Theorem 1.2. Write

$$P_{56} = P \cap \bigcap_{i=1}^{56} \{z :\ \langle z,u_i\rangle \leq 2/\sqrt{3}\}$$

for the cell of Theorem 1.2.

Lemma 5.2. $P \cap \operatorname{conv}(M) = P_{56}$.

Proof. The inclusion $\subseteq$ is Lemma 3.3 applied to the deep holes. For $\supseteq$ we must show that $\langle z,s\rangle \leq 2$ for every $z \in P_{56}$ and every shallow hole s. The Weyl group $W(E_7)$, of order 2 903 040, preserves Λ, M, P and the set of deep holes, hence P_{56}, and it acts transitively on the 576 shallow holes (for finite group actions of this kind and the equivariant bookkeeping they allow, compare [Bha22, Bha23]) (the stabiliser of one of them is $W(A_6)$, of order $5040 = 2\,903\,040/576$). So

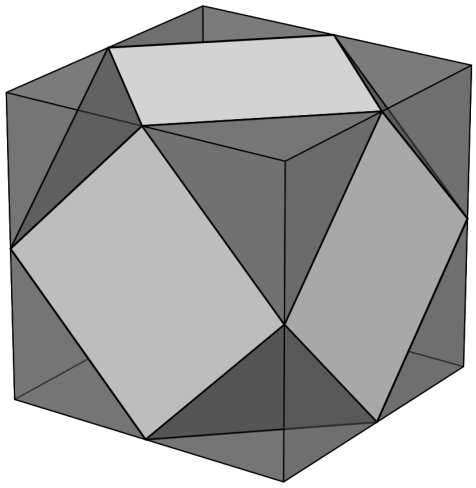


FIGURE 6. The shape of Theorem 1.5, in the three-dimensional analogue $\mathbb{Z}^3$: the cell (a cube) meets the convex hull of the minimal vectors (an octahedron) in a cuboctahedron (light), and the caps at the eight corners (dark) are exactly what extra centres can remove. As for E_7, every edge of the cell leaves a cut corner, and the caps of two adjacent corners meet at the midpoint of the edge between them. For E_7 there are 56 caps, each of volume $\frac{1}{105}$.

it is enough to maximise the single linear function $z \mapsto \langle z, s\rangle$ over P_{56}, and the maximum is attained at a vertex. The exact enumeration of Section 10 lists the 5364 vertices of P_{56}: the 576 shallow holes of P and 4788 new vertices, one on each edge of P. (Every edge of P leaves a deep hole: 4032 edges join a deep hole to a shallow one and 756 join two deep holes, and on the latter the two cuts meet at the midpoint.) Every one of them satisfies all 632 facet inequalities of $\operatorname{conv}(M)$; in fact the maximum of $\langle z, s\rangle$ over P_{56} is $|s|^2 = \frac{7}{4} < 2$, attained only at $z = s$. □

Proof of Theorem 1.5. By Lemma 5.1, every half-space $H(y)$ with $y \notin \{0\} \cup M$ contains $\operatorname{conv}(M)$, so

$$V_0(\mathcal{C}) = P \cap \bigcap_y H(y) \;\supseteq\; P \cap \operatorname{conv}(M) = P_{56}$$

by Lemma 5.2, and $\operatorname{vol} P_{56} = 232/15$ by Theorem 1.2. The packing of Theorem 1.2 contains $\{0\} \cup M$ and its cell at the origin is P_{56}, so the bound is attained. □

Remark 5.3. Theorem 1.5 is a statement about all packings containing the shell, with any number of further centres at any positions; it does not need the shell bound of Lemma 2.2, the radial structure of Theorem 1.1 or any restriction on how the extra centres interact with each other. The inclusion $V_0 \supseteq P \cap \operatorname{conv}(M)$ holds for every lattice with minimal distance 2 whose relevant vectors are its minimal vectors, and for the lattices that pass the criterion of Theorem 1.6 it says that the lattice cell is the smallest cell around the lattice shell. For E_7 the two bodies differ by the 56 caps, and nothing smaller is possible.

Remark 5.4. Theorem 1.5 does not say that a centre with 126 neighbours at distance 2 has a cell of volume at least 232/15. The kissing number in dimension 7 is not known: $126 \le \tau_7 \le 134$, the upper bound being the semidefinite programming bound of Mittelmann and Vallentin [MV10], and it is not known whether the configuration of 126 points is unique. The theorem applies to the E_7 shell, and to any shell isometric to it.

6. A lower bound for every cell

The upper end of Theorem 1.4 is Theorem 1.2. The lower end is the simplex bound of Rogers [Rog58] in the cell-by-cell form proved by Böröczky [Bör78]. Let T_n be a regular n-simplex of edge length 2 and let Ω_n be the fraction of the full solid angle that T_n subtends at one of its vertices, so that the $n+1$ unit balls centred at the vertices meet T_n in total volume $(n+1)\,\Omega_n \operatorname{vol}(B^n)$. Put

$$\sigma_n = \frac{(n+1)\,\Omega_n \operatorname{vol}(B^n)}{\operatorname{vol}(T_n)}.$$

Theorem 6.1 (Rogers, Böröczky)**.** *Let $\mathcal{C}$ be a unit-ball packing of $\mathbb{R}^n$ and $c \in \mathcal{C}$ a centre whose Voronoi cell is bounded. Then* $\operatorname{vol}(B^n) \le \sigma_n \operatorname{vol} V_c$.

Rogers proved the inequality for the density of a packing by dissecting the Voronoi cells into orthoschemes; Böröczky [Bör78, Theorem 1] showed that the same inequality holds for each cell separately, in spaces of constant curvature. Since $\operatorname{vol}(T_n) = \frac{2^n}{n!}\sqrt{\frac{n+1}{2^n}}$, the theorem says

$$\operatorname{vol} V_c \;\ge\; \frac{\operatorname{vol}(B^n)}{\sigma_n} = \frac{\operatorname{vol}(T_n)}{(n+1)\,\Omega_n}, \qquad \operatorname{vol} V_c \;\ge\; \frac{2/315}{8\,\Omega_7} = \frac{1}{1260\,\Omega_7} \quad (n=7). \tag{6.1}$$

The cell of a centre in a saturated packing is bounded, and an unbounded cell has infinite volume, so (6.1) holds for every cell of every packing.

6.1. **The solid angle.** The cone of T_n at a vertex is spanned by unit vectors $e_1, \dots, e_n$ with $\langle e_i, e_j\rangle = \frac12$ for $i \ne j$. If X is a standard Gaussian vector in $\mathbb{R}^n$, its direction is uniform on the sphere, so Ω_n is the probability that X lies in the cone. Write $X = \sum a_i e_i$; then X lies in the cone if and only if all $a_i \ge 0$. The vector $a = G^{-1}(\langle X, e_i\rangle)_i$, with Gram matrix $G = \frac12(I+J)$, is Gaussian with covariance $G^{-1} = 2\big(I - \frac{1}{n+1}J\big)$, whose correlations are all equal to $-\frac1n$. Hence Ω_n is the orthant probability

$$\Omega_n = \Pr\big(Z_1 \ge 0, \dots, Z_n \ge 0\big), \qquad \operatorname{corr}(Z_i, Z_j) = -\tfrac1n \ (i \ne j).$$

For equicorrelated Gaussian vectors with correlation $\rho \ge 0$ one has the classical representation (Tong [Ton90, Section 8.2]) $\Pr(Z \ge 0) = \int_{\mathbb{R}} \varphi(t)\,\Phi\big(t\sqrt{\rho/(1-\rho)}\big)^n\,dt$, with φ and Φ the standard normal density and distribution function. Both sides are analytic in ρ on $(-\frac{1}{n-1}, 1)$, and continuing to $\rho = -\frac1n$, where $\sqrt{\rho/(1-\rho)} = i\,a$ with $a = (n+1)^{-1/2}$, gives

$$\Omega_n = 2\int_0^\infty \varphi(t)\,\operatorname{Re}\Big[\Big(\tfrac12 + \tfrac{i}{2}\operatorname{erfi}\big(\tfrac{at}{\sqrt2}\big)\Big)^n\Big]\,dt, \tag{6.2}$$

since $\Phi(ix) = \frac12 + \frac{i}{2}\operatorname{erfi}(x/\sqrt2)$. The integrand is $O\big(e^{-t^2/(2(n+1))}\big)$, so the integral converges absolutely. Formula (6.2) gives $\Omega_2 = \frac16$ and $\Omega_3 = 0.0438699\ldots$, the solid angle of a regular tetrahedron, as it should, and

$$\Omega_7 = 5.5409353698599424466\ldots \times 10^{-5}, \qquad \frac{1}{1260\,\Omega_7} = 14.32340824561638\ldots$$

These values were computed from (6.2) at 40 significant digits, in two independent implementations, and checked against a direct numerical evaluation of the Gaussian orthant probability (Section 10). The corresponding centre density $1260\,\Omega_7 = 0.06981578\ldots$ is the value of Rogers' bound listed by Conway and Sloane [CS99, Table 1.2].

Proof of Theorem 1.4. The lower bound is (6.1) with the value of Ω_7 above; the upper bound is Theorem 1.2. Finally $\operatorname{vol}(B^7)/14.3234 = (16\pi^3/105)/14.3234 < 0.3299$. □

Remark 6.2. The lower bound is what a cell would have if all the Delaunay simplices at its centre were regular simplices of edge 2, which cannot happen in $\mathbb{R}^7$ since regular simplices do not fill space around a point; the upper bound comes from an explicit packing. The window [14.3234, 15.4667] is therefore wide for a structural reason, and narrowing it from below would require a lower bound that knows that regular simplices do not tile.

TABLE 1. The solid angle Ω_n at a vertex of the regular simplex, the cell bound $\mathrm{vol}(T_n)/((n+1)\Omega_n)$ of (6.1) and the cell of the densest known lattice, both for unit balls.

n	Ω_n	cell bound	lattice cell
2	1/6	3.4641	$2\sqrt{3} = 3.4641$
3	0.0438699	5.3727	$4\sqrt{2} = 5.6569$
4	0.0097847	7.6176	8
5	0.0019220	10.0128	$8\sqrt{2} = 11.3137$
6	3.40695×10^{-4}	12.3266	$8\sqrt{3} = 13.8564$
7	5.54094×10^{-5}	14.3234	16
8	8.36884×10^{-6}	15.8057	16

7. The other root lattices

We now run the criterion of Theorem 1.6 through the irreducible root lattices. In the root normalisation (roots of squared length 2) the covering radii are classical [CS99, Ch. 4 and Ch. 21]:

$$\rho_0(A_n)^2 = \frac{a(n+1-a)}{n+1} \ \ (a = \lfloor \tfrac{n+1}{2} \rfloor), \qquad \rho_0(D_n)^2 = \frac{n}{4} \ (n \geq 4),$$
$$\rho_0(E_6)^2 = \frac{4}{3}, \qquad \rho_0(E_7)^2 = \frac{3}{2}, \qquad \rho_0(E_8)^2 = 1.$$

Scaling to minimal distance 2 multiplies squared lengths by 2, so $\rho^2 = 2\rho_0^2$, and the necessary condition $\rho \leq \sqrt{2}$ of Theorem 1.6 reads $\rho_0 \leq 1$. The table lists ρ^2 for the lattices of Theorem 1.7; it is also what the vertex enumeration of Section 10 finds as the largest squared vertex norm.

Λ	A_1	A_2	A_3	D_4	E_8	A_4	A_5	D_5	D_6	D_7	E_6	E_7
ρ^2	1	$\frac{4}{3}$	2	2	2	$\frac{12}{5}$	3	$\frac{5}{2}$	3	$\frac{7}{2}$	$\frac{8}{3}$	3

So A_n for $n \geq 4$, D_n for $n \geq 5$, E_6 and E_7 fail the criterion outright, by the last sentence of Theorem 1.6, and it remains to prove the inclusion $P \subseteq \mathrm{conv}\, M$ for A_1, A_2, A_3, D_4 and E_8, and to compute the fractions in the table of Theorem 1.7.

7.1. **The lattices that pass.** For A_1, A_2, A_3, D_4 and E_8 the table gives $\rho \leq \sqrt{2}$, so $P \subseteq \mathrm{conv}\, M$ by Theorem 1.6; the relevant vectors of these lattices are their minimal vectors (Conway and Sloane [CS99, Ch. 21]; for A_3 and D_4 this is also confirmed by Lemma 2.5 and the exact volumes of Section 10). The inclusion is visible in small cases. In the root normalisation the cell of D_4 is the 24-cell with vertices $\pm e_i$ and $\frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1)$, each the midpoint of two orthogonal roots, for example $\frac{1}{2}(1,1,1,1) = \frac{1}{2}((e_1+e_2)+(e_3+e_4))$, so here P touches the boundary of $\mathrm{conv}\, M$ at every vertex, as $\rho = \sqrt{2}$ requires. For E_8 the deep hole e_1 is the midpoint of $e_1 \pm e_2$, and the shallow hole $h = (-\frac{5}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6})$ is the average of the roots $-e_1 + e_8$, $-e_1 - e_8$ and $\frac{1}{2}(-1,1,-1,-1,-1,-1,-1,1)$; since $\langle h, v' \rangle \leq |h|\,|v'| \leq \sqrt{8/9} < 1$ for every vertex v', it even lies in the interior of $\mathrm{conv}\, M$ (in the root normalisation $\mathrm{conv}\, M$ is cut out by $\langle x, v' \rangle \leq 1$).

7.2. **The lattices that fail, and how much they lose.** For each of A_4, A_5, D_5, D_6, D_7, E_6 and E_7 the computation of Section 10 enumerates the vertices of the cell Q cut out by the roots, confirms $\mathrm{vol}\, Q = \mathrm{covol}$ (so $Q = P$ by Lemma 2.5), picks a vertex v of largest norm, and computes exactly the volume of $P \cap \{w : \langle w, v \rangle \leq 1\}$ in the root normalisation. The half-space $\langle w, v \rangle \leq 1$ is the bisector of the extra centre $(2/\rho_0^2)v$, which is at distance exactly $\sqrt{2}$ (the minimal distance in this normalisation) from the roots through v and farther from the others; this is the configuration of Theorem 1.6, scaled. The ratio $\mathrm{vol}(P \cap \{\langle w, v \rangle \leq 1\})/\mathrm{vol}\, P$ is a rational number independent of the scale, and one minus this ratio is the fraction in the table of Theorem 1.7. For E_7 it is $\frac{1}{1680}$, and $16 \cdot \frac{1}{1680} = \frac{1}{105}$ is Proposition 4.9 again, computed in a

third way (in the root normalisation rather than the Λ-picture). This completes the proof of Theorem 1.7. □

Remark 7.1. The fractions are small, which is as it should be: a single ball placed in a deep hole direction trims one corner of the cell. For E_7 the corner is a cone over a polar 2_{21} and the fraction has the clean value $\frac{1}{1680} = \frac{1}{7\cdot 240}$; for D_7 the corner at the deep hole $\frac{1}{2}(1,\dots,1)$ is a cone over a polytope with less symmetry, and the fraction $\frac{817}{201600}$ is correspondingly less tidy. We have not pursued the multi-hole versions of the construction for the other lattices.

8. Where the earlier argument fails

Versions 1–9 of this paper deduced $\Delta_7 = \pi^3/105$ from the statement that every Voronoi cell of a unit-ball packing of $\mathbb{R}^7$ has volume at least 16. The deduction itself is Proposition 2.3; the statement is false by Theorem 1.1. For the reader who has the earlier versions we identify the step that fails and the steps that, while not fatal, do not hold as stated.

8.1. **The reduction to neighbours at distance 2.** The argument first moved every active neighbour $y_i = r_i u_i$ radially to $2u_i$, using that the cell grows when a single neighbour is moved outwards. That monotonicity is correct, and it points the right way: moving a neighbour inwards shrinks the cell, so the reduced configuration has a smaller cell and a lower bound for it would be a lower bound for the original one. The difficulty is that the reduced configuration need not be a packing. If $\langle u_i, u_j\rangle > \frac{1}{2}$ then $|2u_i - 2u_j| < 2$. This happens in Theorem 1.1: the extra centre has direction u with $\langle u, m/2\rangle = 1/\sqrt{3} > \frac{1}{2}$ for each of the 27 minimal vectors m that it touches.

8.2. **The overlapping-cap step.** To deal with such pairs, the earlier versions argued that when $\langle u_i, u_j\rangle > \frac{1}{2}$ the centre y_j is *not a valid packing centre simultaneous with* y_i, so that its half-space may be omitted, and concluded

$$V_0 \;\supseteq\; P \cap \{z :\; \langle z, u_i\rangle \le 1\}.$$

Both halves of this step fail. The centres y_i and y_j are centres of the original packing, at distance at least 2 from each other; it is only their images $2u_i, 2u_j$ that collide, and the images are not centres of anything. And omitting a half-space from the intersection that defines V_0 enlarges the body, so the displayed inclusion goes the wrong way: the true cell is contained in the right-hand side, not the other way round, and no lower bound for $\operatorname{vol} V_0$ can be derived from it. In the packing of Theorem 1.1, with y_i any of the 27 touched minimal vectors and y_j the extra centre, the inclusion would give $\operatorname{vol} V_0 \ge 16$, whereas the true volume is $16 - \frac{1}{105}$.

8.3. **Local positivity.** The earlier versions also proved that the Hessian of the cell volume, as a function of the directions of 126 neighbours at distance 2, is positive definite at the E_7 configuration, with smallest pivot $\frac{823}{3360}$ in a rational $LDL^\top$ factorisation. This is a local statement about configurations of 126 neighbours on the contact sphere, near the E_7 configuration. We have not re-derived it, and we do not use it. Whatever its status, it cannot bear on Theorem 1.1, whose extra centre is not on the contact sphere, and the same applies to the exchange certificates computed for 115 chamber types, which also concern neighbours at distance 2.

8.4. **Two further points.** First, the earlier versions used $\sqrt{3}$, the circumradius of P, as the circumradius of every cell of every packing. In a saturated packing the correct bound is 2 (Section 2), and with that change the density argument of Proposition 2.3 goes through; the defect is harmless. Second, they described a Steiner symmetrisation of the Voronoi cell of $\sqrt{2}\,E_7$ as a deformation of lattices. A Steiner symmetral of a Voronoi cell is in general not the Voronoi cell of any lattice, so this deformation does not exist as described; it was not used in the main line of the argument.

TABLE 2. The steps of versions 1–9 and their status.

step	status
cells bound density (Proposition 2.3)	correct, with circumradius 2
radial monotonicity of the cell	correct
reduction to neighbours at distance 2	reduced configuration need not be a packing
omitting the half-space of y_j	enlarges the cell; inclusion reversed
Hessian positivity at E_7	local only; not used here
Steiner deformation	not a deformation of lattices; not needed
every cell has volume ≥ 16	false (Theorems 1.1 and 1.2)
$\Delta_7 = \pi^3/105$	open

9. CONSEQUENCES FOR THE SPHERE PACKING PROBLEM IN DIMENSION 7

9.1. **What is known.** The lattice E_7 is the densest lattice in dimension 7 (Blichfeldt [Bli35]; see [CS99, Ch. 1] for the history), with density $\pi^3/105 = 0.295297\ldots$. Whether it is the densest packing of all is open. The best upper bound comes from the linear programming method of Cohn and Elkies [CE03], which gives 0.327456 in dimension 7; Rogers's simplex bound [Rog58] gives 0.3299. The gap between the lower and the upper bound is about 11% of the lower bound, far larger than in dimensions 8 and 24, where the linear programming bound is sharp [Via17, CKMRV17, Coh17].

9.2. **What the local approach can and cannot do.** Proposition 2.3 describes the one local method that needs no weights: bound each cell from below. By Theorem 1.4 the best such bound in dimension 7 is a number ω_7 between 14.3234 and 232/15, and the method proves exactly $\Delta_7 \leq \mathrm{vol}(B^7)/\omega_7$. With the known lower end this is Rogers' bound 0.3299; even if ω_7 were equal to 232/15, the method would give only $\Delta_7 \leq \frac{15}{232}\,\mathrm{vol}(B^7) = 0.3054\ldots$. That would improve the linear programming bound 0.3275 but would still not reach $\pi^3/105 = 0.2953$. Determining ω_7 is therefore a worthwhile problem in its own right, even though it cannot settle Δ_7. Let us be precise about what the examples refute.

(1) The statement *every Voronoi cell of a unit-ball packing of* $\mathbb{R}^7$ *has volume at least* 16 is false, for finite, for saturated and for periodic packings alike (Theorem 1.1, Corollary 1.3).
(2) The statement remains false if one restricts to packings in which some centre has the full E_7 contact configuration, since the examples have it; for those packings the correct bound is 232/15 (Theorem 1.5).
(3) The statement cannot be rescued by moving the neighbours to the contact sphere: in Theorem 1.1 the 126 neighbours are already on it, and the extra centre, which is not on it, is the whole point. An argument that reduces a general configuration to one in which every neighbour is at distance 2 must fail, since after that reduction the extra centre would violate the packing constraint with the 27 balls it touches, and the reduced configuration is not a packing (Section 8).

9.3. **What survives.** Two positive statements about the E_7 cell survive and are worth recording. First, the cell P is cut out by the minimal vectors alone (Lemma 2.5 with $\mathrm{vol}\,P = 16$), so any configuration whose active neighbours are a subset of the E_7 contact configuration has a cell containing P, of volume at least 16. Second, the extra centres of Theorems 1.1 and 1.2 remove volume from the cell of the origin but have large cells themselves; the deficit is local and is compensated nearby. This is the behaviour that any proof of $\Delta_7 = \pi^3/105$ along the lines of the Kepler proof would have to quantify: assign to each centre a weighted average of the volumes of its cell and its neighbours' cells, and bound the average from below by 16. We do not know such weights.

9.4. **Questions.**

Question 9.1. *What is ω_7, the infimum of* $\operatorname{vol} V_c$ *over all centres c of all unit-ball packings of $\mathbb{R}^7$? Theorem 1.4 gives* $14.3234 < \omega_7 \le 232/15$*, and Theorem 1.5 shows that the upper end is attained among cells whose centre carries the E_7 shell.*

Question 9.2. *Does the Voronoi cell of the Leech lattice lie in the convex hull of its minimal vectors? Its relevant vectors are not only its minimal vectors, so Theorem 1.6 does not apply directly, although the ratio of covering radius to packing radius is again $\sqrt{2}$* [CPS82]*. If the inclusion holds, the Leech lattice joins A_1, A_2, A_3, D_4 and E_8 as a lattice for which no single extra centre shrinks the cell.*

Question 9.3. *Is there a weighted local bound in dimension* 7*, in the sense of Hales, that proves $\Delta_7 = \pi^3/105$? The examples of this paper constrain the weights: in the* 183*-point configuration of Theorem 1.2 the weight of the origin's cell must be compensated by the cells of the* 56 *extra centres and of the balls that saturate the packing.*

10. Verification

All computations the proofs rest on are finite and exact. They are carried out by two sets of programs, in Julia and in Lean 4, which are archived with the paper (Section 10.4); this section says what they compute, so that the computations can be repeated with any other software.

10.1. **Exact polytope computations.** A polytope is given by facet inequalities with rational coefficients in a rational chart of the subspace that contains it. Its vertices are computed by the double description method in the implementation `cddlib` of Fukuda and Prodon [FP96], in its GMP rational mode, through the Julia package `Polyhedra`; volumes are computed exactly from a triangulation, and a volume in the chart is converted into a true volume by the square root of the Gram determinant of the chart. For E_7 the chart is $(x_1, x_3, \dots, x_8)$ on the plane $x_1 + x_2 = 0$ in the root normalisation; its Gram determinant is 2, and 7-volumes in the Λ-picture are 16 times chart volumes. The following are checked, each as an exact equality of rational numbers.

(1) *The cell.* The 126 inequalities $\langle w, \alpha\rangle \le 1$ define a polytope with 632 vertices, 56 of squared norm $\frac{3}{2}$ and 576 of squared norm $\frac{7}{8}$; its chart volume is 1, so its volume is $\sqrt{2} = \operatorname{covol} L$, and 16 in the Λ-picture. Each of the 56 deep holes lies on exactly 27 facets. (Lemmas 2.5, 4.2 and 4.5.)
(2) *One cut, two ways.* Adding $\langle w, v_L\rangle \le 1$ gives a polytope with 730 vertices whose volume is $16 - \frac{1}{105}$. Independently, the cap is the convex hull of the deep hole and the 99 vertices of the cross-section (no other vertex of P lies beyond the cut), and its volume computed directly from these 100 points is $\frac{1}{105}$. (Proposition 4.9.)
(3) *The seventh-power law.* For the thresholds $c = \frac{11}{10}, \frac{5}{4}, \frac{7}{5}$ in place of 1, the cap has volume exactly $\frac{1}{105}(3 - 2c)^7$. (Confirms Lemma 4.8 and Proposition 4.9.)
(4) *All* 56 *cuts.* For each deep hole the cap is a cone of volume $\frac{1}{105}$, and every vertex of it satisfies the inequalities of the other 55 cuts, so the caps have disjoint interiors and $\operatorname{vol} P_{56} = 16 - \frac{56}{105} = \frac{232}{15}$. (Theorem 1.2.)
(5) $P_{56} \subseteq \operatorname{conv} M$. P_{56} has 5364 vertices, and each satisfies the 632 inequalities $\langle x, v\rangle \le 1$, v a vertex of P, that cut out $\operatorname{conv} M$ by Lemma 3.3; among the vertices of P exactly the 56 deep holes violate them. For each of the 576 shallow holes s the largest value of $\langle x, s\rangle$ on P_{56} is $|s|^2 = \frac{7}{8} < 1$, taken only at $x = s$. (Lemma 5.2 and Theorem 1.5.)
(6) *The other root lattices.* The same computation for A_3, D_4, A_4, A_5, D_5, D_6, D_7 and E_6, with the roots of A_n in the sum-zero hyperplane of $\mathbb{R}^{n+1}$, of D_n in $\mathbb{R}^n$ and of E_6 as the roots of E_8 orthogonal to $e_1 - e_2$ and $e_2 - e_3$: in each case the chart volume times the square root of the Gram determinant equals the covolume $\sqrt{\det}$, the largest squared vertex norm equals ρ_0^2, and the cut at a deep hole removes the fraction listed in Theorem 1.7. For

E_8 the vertex enumeration gives the 19 440 vertices, 2160 of squared norm 1 and 17 280 of squared norm $\frac{8}{9}$. (Theorem 1.7.)

10.2. **The solid angle.** Ω_n is computed from (6.2) in 256-bit floating point arithmetic: erfi by its power series, which has positive terms, and the integral by adaptive Gauss–Kronrod quadrature on $[0, 80]$ with relative tolerance 10^{-45}; the omitted tail beyond 80 is below e^{-300}. The values for $n = 2, \dots, 8$ are those of Table 1. As an independent check, the orthant probability for $n = 7$ was also computed by the randomised quasi-Monte Carlo method of Genz for multivariate normal probabilities, which agrees with (6.2) to the accuracy of that method: with requested absolute error 10^{-10} it returns 5.540931×10^{-5}.

10.3. **Kernel-checked combinatorics.** The combinatorial facts behind Theorems 1.1 and 1.2 are checked in Lean 4, in integer arithmetic on doubled coordinates, with proofs by evaluation (`native_decide`, which relies on Lean's compiled evaluator as well as its kernel): L has 126 roots, all of squared norm 2; the 56 vectors listed in Appendix A have squared norm $\frac{3}{2}$, lie in P_L and each lies on exactly 27 facets; their pairwise inner products are $\frac{1}{2}$, $-\frac{1}{2}$ or $-\frac{3}{2}$ (Lemma 4.11); the extra centre $\frac{4}{3}v_L$ is at distance at least $\sqrt{2}$ from the origin and from every root, with equality for exactly 27 roots; two extra centres are at distance at least $\frac{8}{3\sqrt{2}}$; and the arithmetic $16 - \frac{56}{105} = \frac{232}{15}$, $16 \cdot 105 = 1680$.

10.4. **Data and code availability.** The Julia and Lean programs and their outputs are in the repository `https://github.com/creelie/voronoi-d7`. No other data were generated or analysed.

Author contributions

Deep Bhattacharjee conceived the study, found the counterexample and the shell theorem, carried out the proofs and the computations, and wrote the paper. Ushashi Bhattacharya checked the proofs and the computations. Shounak Bhattacharya took part in the discussions and checked the manuscript. All authors read and approved the final version.

Declarations

Funding. The work was carried out with the authors' own means; it received no external funding.

Competing interests. The authors declare that they have no competing interests, financial or otherwise, and that no author, institution or affiliation stands to gain or lose from the results.

Ethics. The work involves no human participants, animals or personal data, and raises no ethical issues.

Data availability. All code and outputs are available as stated in Section 10.4.

Appendix A. Coordinates

All vectors are in the root normalisation (the L-picture), in the coordinates of $\mathbb{R}^8$ with $x_1 + x_2 = 0$.

A.1. **The twenty-seven roots through the deep hole v_L.** With $v_L = -\frac{1}{2}(0, 0, 1, 1, 1, 1, 1, 1)$ the set S consists of

$$-e_i - e_j \quad (3 \le i < j \le 8)$$

and

$$(\tfrac{1}{2}, -\tfrac{1}{2}, x_3, \dots, x_8), \qquad (-\tfrac{1}{2}, \tfrac{1}{2}, x_3, \dots, x_8),$$

where $x_p = \frac{1}{2}$ for one $p \in \{3, \dots, 8\}$ and $x_i = -\frac{1}{2}$ for $i \neq p$.

A.2. **The fifty-six deep holes.** The 32 vectors $\frac{1}{2}(0,0,\varepsilon_3,\dots,\varepsilon_8)$ with $\varepsilon_i=\pm1$ and an even number of plus signs, and the 24 vectors $(\frac12,-\frac12,\pm e_i)$, $(-\frac12,\frac12,\pm e_i)$ for $3\le i\le 8$. They come in 28 antipodal pairs, and by Lemma 4.11 each has 27 others at inner product $\frac12$, 27 at $-\frac12$ and one at $-\frac32$; the distances between the corresponding extra centres of Theorem 1.2 are $\frac{8}{3}$, $\frac{8\sqrt{2}}{3}$ and $\frac{8}{\sqrt{3}}$.

A.3. **A shallow hole.** The vertex $s=(\frac14,-\frac14,-\frac12,-\frac12,-\frac12,0,0,0)$ has squared norm $\frac78$ and lies on the seven facets of the roots $-e_3-e_4$, $-e_3-e_5$, $-e_4-e_5$ and $(\frac12,-\frac12,-\frac12,-\frac12,-\frac12,x_6,x_7,x_8)$ with (x_6,x_7,x_8) one of $(\frac12,\frac12,\frac12)$, $(\frac12,-\frac12,-\frac12)$, $(-\frac12,\frac12,-\frac12)$, $(-\frac12,-\frac12,\frac12)$. The Delaunay cell at s is the simplex with vertices 0 and these seven roots, all at distance $\sqrt{7/8}$ from s, and every other root α has $\langle s,\alpha\rangle<1$. All 576 shallow holes lie in the interior of $\operatorname{conv} M$ (the vertex enumeration of Section 10 confirms this for each of them), which is why only the deep holes enter Theorem 1.6 for E_7.

A.4. **The distances in Theorem 1.1 at a glance.** In the Λ-picture (minimal distance 2): $|v|=\sqrt3=1.732\dots$; the extra centre at $|y|=4/\sqrt3=2.309\dots$; its bisector at distance $2/\sqrt3=1.154\dots$ from the origin; the cap of height $1/\sqrt3=0.577\dots$; the cross-section at that height is $\frac{1}{\sqrt6}\Pi$, with circumradius $\frac{2}{\sqrt6}=0.816\dots$ and 6-volume $\frac{\sqrt3}{15}=0.1154\dots$; the cap has volume $\frac{1}{105}=0.009523\dots$

Appendix B. Index of notation

$\mathcal{C}$	a unit-ball packing (centres at mutual distance ≥ 2)
$V_c(\mathcal{C})$	Voronoi cell of the centre c
Λ, P, M	lattice with minimal distance 2, its cell at 0, its minimal vectors
L	the copy of E_7 in E_8 orthogonal to $a=e_1+e_2$; $\Lambda=\sqrt{2}L$
$u_m=m/2$, $u_\alpha=\alpha/\sqrt2$	unit normals of the facets of P
ρ, ρ_0	covering radius in the Λ- and L-pictures
v, v_L, u	the deep hole (4.2) and its direction
S	the 27 roots through v_L
C_v, K_t	the cone at v and the cap of height t
Π, n_α	the polar of 2_{21} and the 27 unit vectors of Lemma 4.6
h_K	support function of K
$y_r=ru$	the extra centre at distance r

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